

North American Bat Monitoring Program in Alberta

Data Summary and Activity Trend Analyses: 2019 - 2025

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within Alberta, with key support from the provincial government.

Partners in Alberta have been collecting NABat data since 2015, and as of 2025 there are 36 grid cells that have been sampled more than 3 years, and are considered long term monitoring sites, while there are 16 grid cells with enough repeat sample data (more than 5 years) that could be used for trend modelling. Trend estimates reveal...

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). NABat uses a spatially-balanced random sampling design and standardized protocols for conducting short-term acoustic surveys and colony counts (Loeb et al. 2015).

NABat relies on regional hubs to help coordinate and standardize monitoring across smaller areas. Regional hubs help facilitate local implementation of NABat and maximize the benefits of the program within each region. The North of Northwest (NNW) Bat Hub was established in 2024 with collaboration from the governments of Alaska, Alberta, and British Columbia. The hub was formed by combining two previous regional hubs: the Alberta Hub and the BC and Southeast Alaska Hub. The NNW Bat Hub is directed by a steering committee consisting of representatives from each jurisdiction, major stakeholders, and Government of Alberta representatives and it is coordinated by Biodiversity Pathways. The NNW Hub is responsible for compiling data, identifying calls to species, conducting statistical analyses, and generating annual regional reports. NABat monitoring in Alberta is implemented by many partners including Alberta Parks, National Parks, ACA and MNA.

2.2 Threats to bat populations in Alberta

North American bat populations are currently facing multiple anthropogenic threats, including logging and other forms of habitat loss and fragmentation, disease, pesticides, wind energy development, and climate change. Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns for Albertan bat populations.

2.3 White-Nose Syndrome

White-nose syndrome (WNS) is one of the most wide spread and deadly threats to bats in Alberta. WNS is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (Pd) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta, where it has already caused a significant number of mortality in Little Brown bats across known hibernacula (Wilkinson pers. comm.; Figure 1: Segers (n.d.)).

The impacts of white-nose syndrome have been most severe east of the Rocky Mountains, where bats hibernate in large aggregations in caves and mines, resulting in high mortality (Cheng et al. 2021). In parts of the Rocky Mountains and further west, Little Brown *Myotis* hibernate singly or in small groups in holes in the soil or between rocks in talus deposits (Neubaum 2018; Blejwas et al. 2021) which may reduce mortality relative to eastern North America, but will also make it difficult to detect the spread of WNS. Given the cost and difficulty of monitoring bats in Alberta using traditional methods like capture and colony counts, it is important to use alternative methods for monitoring bats that do not require capture to assess changes or impacts from WNS or other threats.

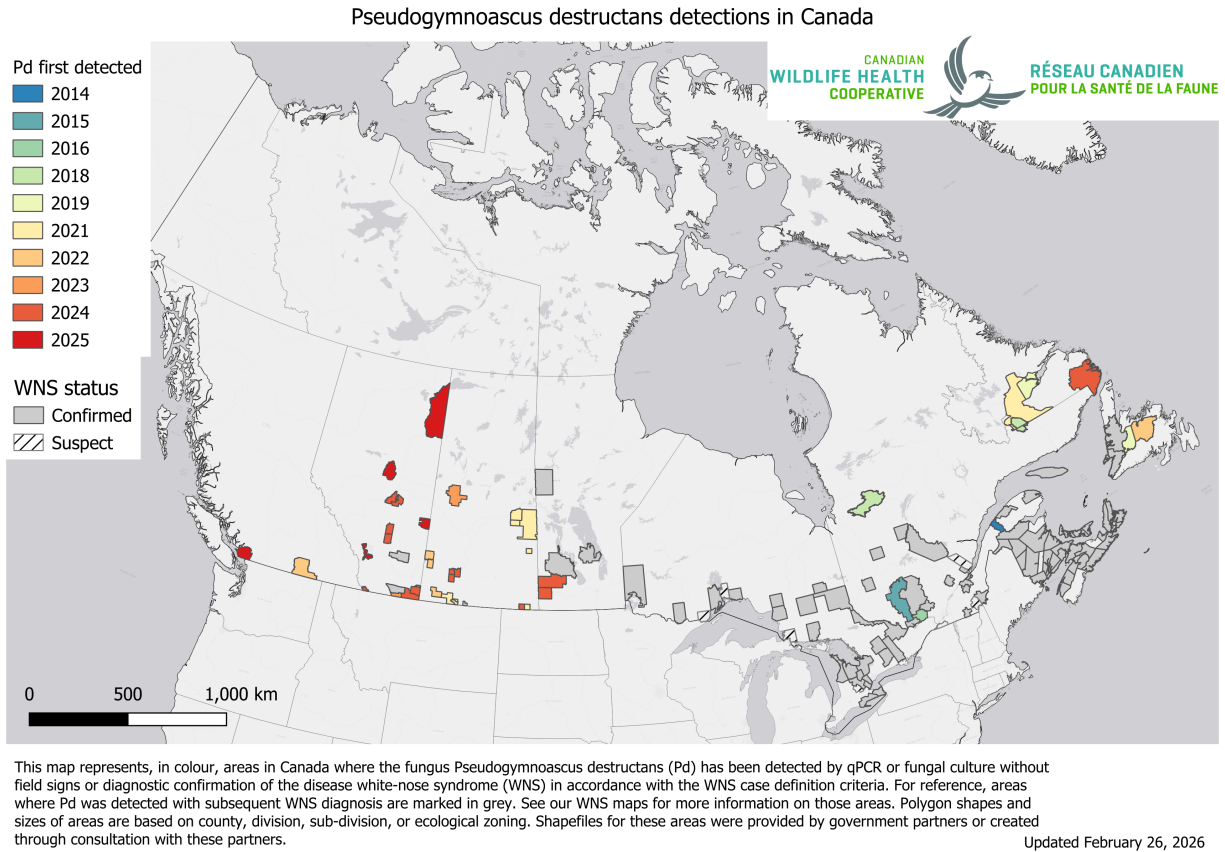


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in Canada. Source: CWHC-RCSF (2026)

2.3.1 Wind Energy Development

Wind energy farms has been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities (Zimmerling and Francis 2016). In Alberta alone, one estimate predicts an annual mortality rate of 10.9 bats per wind turbine (Zimmerling and Francis 2016). There are currently 1778 turbines active across the province (49 different wind farms) with the most turbines found in the grasslands (Figure 2). Based on Zimmerling and Francis (2016) this could mean a total loss of over 19,000 bats per year in the province. As of 2023, COSEWIC designated all three of Alberta's long distance migratory species as endangered due in part to fatalities at wind turbine facilities Canada and Change (2024).



Figure 2: Current wind turbines in Alberta based on the Canadian Wind Turbine Database (Source: <https://search.open.canada.ca/openmap/79fdad93-9025-49ad-ba16-c26d718cc070>)

2.4 Acoustic Monitoring of Bats

Acoustic recordings are a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data and trend estimates cannot be produced for all species using acoustic methods alone:

1. **Imperfect species identification** — Each bat species can produce a broad range of echolocation calls, corresponding to different behaviours in response to environmental features. Species with similar ecology and physiology often produce overlapping calls, particularly in areas with high clutter (obstacles), such as forested habitats. Although detectors are deployed to maximize recordings of diagnostic (typically Search Phase calls), species identifiability will always vary depending on the local species assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, whereas species with quieter calls or specialized habitat use may be less reliably recorded.
3. **Data variability** — Many factors influence recording quality and bat activity, including weather, prey abundance and distribution, and local habitat. Additionally, stationary detectors may record an individual bat multiple times, meaning stationary acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets, but is known to have high misclassification rates for bats. Manual verification generally produces more accurate results, but has its own limitations, including significantly higher processing time and cost. We use a combination of both approaches, as detailed in the [Manual Verification](#) subsection of the Methods.

2.5 Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

3 Methods

3.1 Data Collection

3.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites were selected to maximize species detection and recording quality across different habitat types following NABat guidance (Loeb et al. 2015). Deployment occurred between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing were kept as consistent as possible between years to minimize variability in conditions affecting bat activity.

3.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Bats were recorded along the road by driving at ~30 km/h with a microphone mounted on the vehicle's roof. Transects were 25-35km long and were surveyed twice during the stationary deployment period whenever possible.

3.1.3 Landscape and Weather Covariates

We considered the same covariates for the activity trend models as those previously established in British Columbia (Rae and Lausen 2024). Temperature and relative humidity were drawn from the nearest climate station. Site-level covariates were derived from GIS (elevation, distance to water, water body type) or recorded by grid leaders during deployment (distance to clutter, percent clutter).

We explored additional factors influencing colonization and extinction in the dynamic occupancy models. For those models we included distance to the nearest road, and distance to timber harvest, calculated as the distance from each survey location to the closest road or harvest feature. Road and harvest features were derived from the Human Footprint Index (HFI) published by ABMI.

3.1.4 Manual Verification

From 2015-2023 all recordings were processed using a classifier created by the province of Alberta. This classifier has since been depreciated. 2024 and 2025 acoustic analysis followed the standards set by WCSC in previous years in British Columbia (Rae and Lausen 2024). This included running all files through two automatic classifiers, Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018) and previous analysis standards developed by WCSC (Rae and Lausen 2024). The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually

verified. Instead, based on findings from Stratton et al. (2022) , at least 50% of recordings were selected for manual verification.

3.1.4.1 *Subsetting for manual verification*

Given the project’s large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and Brian Paterson, developed procedures to ensure that selecting a subset of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of recordings that received an autoID tag at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years’ identifications) and commonly misidentified species were fully vetted. Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and “NoID” recordings were randomly selected until 50% of all recordings were reviewed.

3.2 Statistical Analyses

3.2.1 Activity Trends

We conducted species-specific trend analyses using negative binomial GLMM models as described in Rae and Lausen (2024) We restricted the analysis to data from 2019 onward, due to the low number of grid cells that had been sampled in the years prior. We further limited the analysis to grid cells with at least five years of sampling, since estimating a temporal trend requires enough repeat visits per cell to distinguish a directional signal from year-to-year variation.

All models followed the same basic structure, modelling the total number of detected calls as a negative binomial response with log link:

```
Total detections ~ YearS + log(Sample Nights) + (1|Grid Cell ID) + (1|Quadrant)
+ species specific covariates
```

Where total detections is the total number of detected calls of a given species at a site in a given year. YearS is the standardized survey year (years since 2019) and provides the estimate of the temporal activity trend. log(Sample Nights) is the natural logarithm of the number of detector-nights and accounts for variation in sampling effort among site-years. The random intercepts (1|Grid Cell ID) and (1|Quadrant) allow baseline activity to vary among grid cells and among quadrants within them, respectively.

For each species, we performed model selection by fitting the base model augmented with every combination of up to four covariates, excluding combinations that would introduce collinearity. This produced 387 candidate models per species, from which we selected the model with the lowest AIC as the final covariate set `?@tbl-cov`.

Post-hoc tests were run on the final models to ensure proper model fit. Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years.

3.2.1.1 Use of Automated Identification Results in Stationary Trend Analyses

For stationary trend analyses, species identifications came from the Kaelidoscope Pro AutoID software, not from manual verification. Manual review was used only to remove non-bat noise. Specifically, files that manual review flagged as noise were excluded, but where manual review suggested a different species than the AutoID label, the AutoID classification was retained.

This approach was first identified as necessary by Rae and Lausen (2024), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

3.2.2 Transect Abundance Models

For the transect-based abundance trend models, we used only manually verified detections. When a detection was flagged as a species couplet, it was counted toward each species in the couplet. As above, we restricted the data to transect surveys from 2019 onward and to grid cells with at least five years of sampling.

All models followed the same basic structure, modelling the total number of detected calls as a negative binomial response with log link:

```
Total detections ~ YearS + log(transect length) + (1|Grid Cell ID)
+ species specific covariates
```

Where total detections is the total number of detected calls of a given species at a site in a given year. YearS is the standardized survey year (years since 2019) and provides the estimate of the temporal activity trend. $\log(\text{transect length})$ is the natural logarithm of the total length of the transect sampled and accounts for variation in sampling effort among transects. The random intercept (1|Grid Cell ID) allow baseline activity to vary among grid cells.

Covariates were limited to weather variables and transect length, and candidate models excluded any combination of collinear variables. We fit 59 candidate models per species and selected the one with the lowest AIC as the final covariate set `?@tbl-cov-transects`.

After selection, we ran a series of post-hoc checks to confirm model fit. We treated a species' result as reliable only when these tests agreed that the year effect reflected a genuine regional trend rather than an artefact of sparse counts or a single influential cell.

3.2.3 Dynamic Occupancy

We fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at 49 stationary detector sites located in 19 grid cells in Alberta, as well as 173 stationary detector sites in 56 grid cells in British Columbia (BC). Pooling the two regions allowed the more extensive BC data to inform species-specific detection probabilities and covariate slopes, improving our ability to distinguish true absences from non-detections and thereby sharpening occupancy estimates where Alberta data were sparse. A region effect accounted for structural differences in the landscape between British Columbia and Alberta, allowing us to characterize occupancy trends for Alberta with greater precision than an Alberta-only model would permit.

Data were filtered to the 2019–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using a maximum of 4 nights per site-year. Sites had to have a minimum of 2 nights of data within a given year and data from at least 2 years total. Models were run separately for all 7 bat species.

Initial occupancy (ψ) was modeled as a function of elevation and water nearby, with a random effect for region. Baseline occupancy for the Alberta region was estimated from Alberta detections only and was not forced toward the BC average. Colonization (γ) and persistence (φ) were modeled with time-varying intercepts and distance to road as a covariate. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Julian night.

All continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

Models were fit in a Bayesian framework using JAGS (Version 4.3.2) via the `jagsUI` package (Version 1.6.3) in `r` (Version 4.5.1). We ran 3 chains in parallel for 30,000 iterations each, discarding the first 15,000 as burn-in and thinning the remainder by 5, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normal with mean 0 and precision 0.368 (equivalent to a standard deviation of approximately 1.65), and the standard deviation of the region-level random effect was assigned a `Uniform(0, 5)` prior. Convergence was assessed using `Rhat` diagnostics, with values below 1.1 taken to indicate convergence. Results are summarized using posterior means and 95% Bayesian credible intervals. We calculated an overall occupancy trend metric () as the ratio of estimated occupancy in the final year to that in the initial year; values of > 1 with 95% credible intervals entirely above 1 indicate an increase in occupancy, values < 1 with intervals entirely below 1 indicate a decline, and intervals overlapping 1 indicate an uncertain or stable.

4 Results

As of 2025, there have been 252 NABat grid cells sampled in Alberta (Figure 3). However most of these grid cells were sampled only once or twice, and do not form part of the long term monitoring plan. In total we have 36 grid cells that form part of the long term monitoring effort (with at least 3 years of monitoring data) and a total of 16 that currently have 5 or more years of sample data and can be currently used to estimate trends in the province (Figure 3).

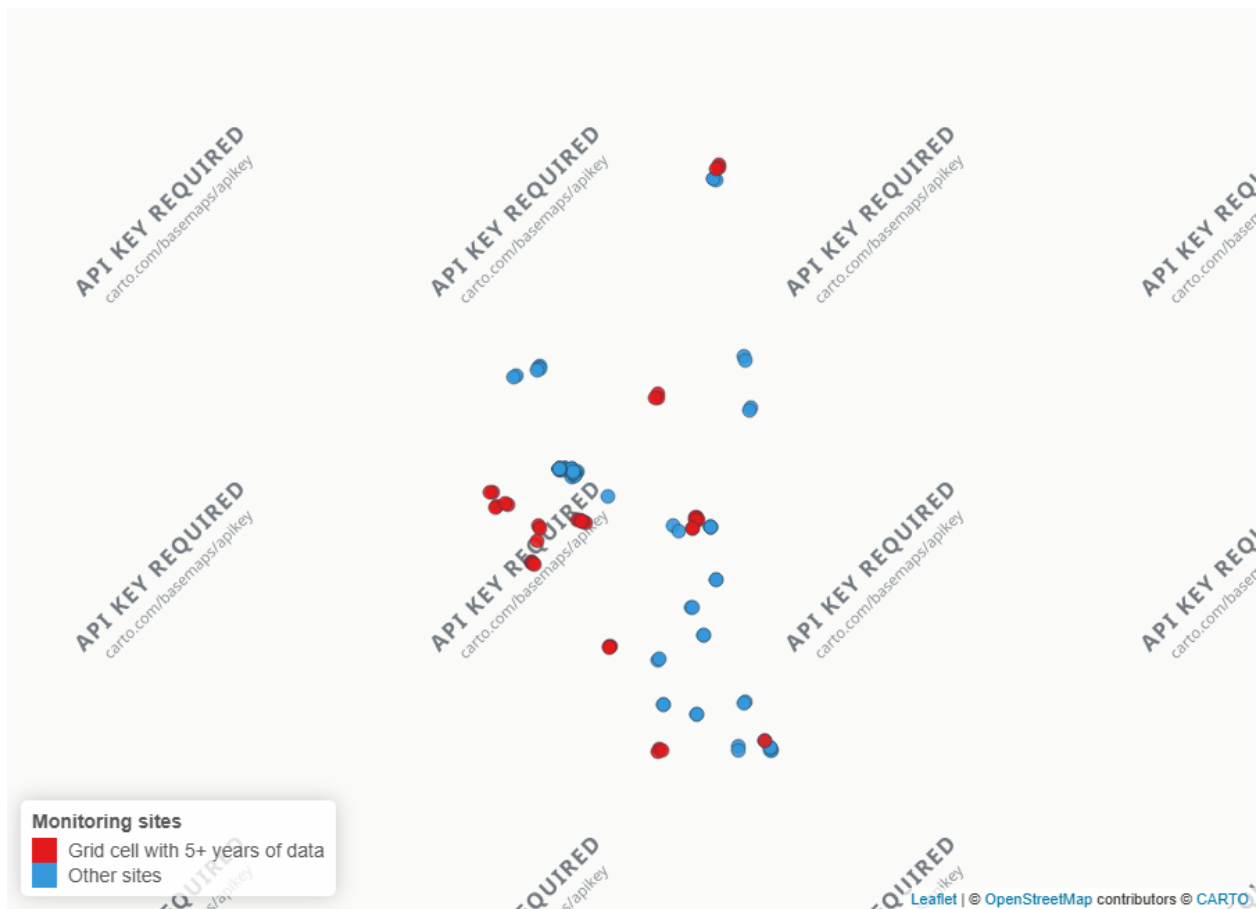


Figure 3: Acoustic bat monitoring sites in Alberta 2015-2025. NABat grid cells monitored for 5 or more years and were used for trend analyses appear in red.

4.1 Activity and Abundance Trends

4.1.1 Covariate Effects

4.2 Dynamic Occupancy

4.2.1 Covariate Effects

5 Discussion

5.1 Looking forward

6 Appendix A

Dynamic occupancy species specific results. Coming soon..

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